

BLUESTOCK FINTECH

Mutual Fund Analytics

Capstone Project I — Final Report

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Bluestock Fintech Internship Program

1. Executive Summary

This report presents the complete analysis of the Bluestock Mutual Fund Analytics Capstone Project. The project analysed **40 mutual fund schemes** across 7 categories using historical NAV data from January 2022 to May 2026. An ETL pipeline was constructed to ingest 10 raw CSV datasets, clean and normalise the data, and load it into a SQLite relational database. Advanced performance metrics including CAGR (1yr, 3yr, 5yr), Sharpe Ratio, Sortino Ratio, Alpha, Beta, VaR (95%), CVaR, Rolling Sharpe, and Sector HHI were computed for all schemes. A composite Fund Scorecard (0–100) was developed based on weighted percentile ranks, and a visual dashboard was produced across 4 analytical pages.

- **Rank 1:** Mirae Asset Large Cap Fund - Regular - Growth
- **Rank 2:** ICICI Pru Midcap Fund - Regular - Growth
- **Rank 3:** Kotak Flexicap Fund - Regular - Growth
- **Rank 4:** HDFC Mid-Cap Opportunities Fund - Regular - Growth
- **Rank 5:** ICICI Pru Bluechip Fund - Direct - Growth

2. Data Sources

The project uses 10 structured CSV datasets provided by Bluestock Fintech, supplemented by live NAV data fetched from the MFAPI (mfapi.in) for 6 key schemes.

#	Dataset	Rows (approx.)	Description
1	Fund Master	40	Scheme metadata, expense ratios, categories
2	NAV History	64,320	Daily NAV for all 40 schemes (Jan 2022–May 2026)
3	AUM by Fund House	Variable	Monthly AUM per AMC
4	Monthly SIP Inflows	~60	Industry-level monthly SIP inflow data
5	Category Inflows	~500	Net inflows per category per month
6	Industry Folio Count	~60	Total active folios across categories
7	Scheme Performance	40	Benchmark-reported return and rating data
8	Investor Transactions	32,778	Synthetic investor SIP/Lumpsum/Redemption records
9	Portfolio Holdings	322	Top stock holdings per equity fund
10	Benchmark Indices	8,050	NIFTY 50 and NIFTY 100 daily closing values

3. ETL Design & Database Architecture

The ETL pipeline was implemented in Python using pandas and SQLite, structured as follows:

3.1 Data Ingestion (Day 1)

- **scripts/data_ingestion.py**: Reads 10 raw CSVs, verifies schemas, creates directory structure.
- **scripts/live_nav_fetch.py**: Fetches live historical NAV via MFAPI REST endpoint.
- **scripts/validate_amfi.py**: Validates AMFI code integrity between fund master and NAV tables.

3.2 Data Cleaning & Transformation (Day 2)

- Deduplication and type coercion on all 10 datasets.
- Date standardisation to ISO 8601 format across all tables.
- Outlier detection in NAV series using IQR-based flagging.
- Categorical normalisation for transaction types and risk categories.

3.3 Database Schema

All cleaned datasets are loaded into a SQLite database (data/db/bluestock_mf.db) with 12 tables following a star-schema pattern. Core fact tables: fact_nav, fact_transactions, fact_aum, fact_performance. Dimension tables: dim_fund, dim_date. Relationship keys: amfi_code (fund identifier), date (time dimension).

4. Exploratory Data Analysis (EDA) Findings

Key observations from the 16 EDA charts generated in Day 3:

- NAV Trends: All equity fund NAVs show a positive trajectory from Jan 2022 to May 2026, with a significant correction in 2022 followed by a sustained bull run.
- Fund Category Distribution: Large-cap and mid-cap funds dominate the tracked universe (50%+), followed by small-cap and debt funds.
- Expense Ratio: Direct plans consistently have ~0.5–0.8% lower expense ratios than regular plans.
- SIP Inflows: Sustained month-on-month SIP inflow growth, reaching ■25,000+ Cr by 2025.
- AUM Concentration: Top 5 AMCs (HDFC, ICICI Prudential, SBI, Mirae Asset, Kotak) account for over 65% of total AUM.
- Investor Demographics: Majority of investors are in the 25–45 age bracket; Tier-1 cities contribute the highest SIP volumes.
- Folio Growth: Industry folio count grew by ~40% over the analysis period, signalling increasing retail participation.
- Correlation Matrix: Large-cap funds show high positive correlation (>0.85) with each other and with Nifty 50; debt funds show near-zero correlation with equity.

5. Fund Performance Analysis

5.1 Fund Scorecard (Top 10)

Scheme Name	Cagr 3Yr (%)	Sharpe Ratio	Sortino Ratio	Max Drawdown (%)	Scorecard
Mirae Asset Large Cap Fund - Regular - Growth	33.99	1.07	1.49	-11.27	86.25
ICICI Pru Midcap Fund - Regular - Growth	31.77	0.88	1.29	-18.19	82.88
Kotak Flexicap Fund - Regular - Growth	29.58	0.97	1.48	-12.97	82.0
HDFC Mid-Cap Opportunities Fund - Regular - Growth	32.43	0.81	1.14	-16.22	80.75
ICICI Pru Bluechip Fund - Direct - Growth	32.48	0.71	1.06	-12.59	79.38
Axis Midcap Fund - Regular - Growth	35.1	0.73	1.06	-20.96	78.25
SBI Bluechip Fund - Regular Plan - Growth	30.45	0.86	1.29	-15.01	74.19
Mirae Asset Tax Saver Fund - Regular - Growth	29.17	0.92	1.35	-16.4	73.69
SBI Small Cap Fund - Regular Plan - Growth	26.66	0.71	1.07	-28.71	68.0
ABSL Frontline Equity Fund - Regular - Growth	28.96	0.72	1.06	-11.29	67.56

5.2 Alpha & Beta — OLS Regression vs NIFTY 100

Alpha and Beta were computed via OLS linear regression (`scipy.stats.linregress`) of each fund's daily returns against NIFTY 100 daily returns. Alpha is annualised ($\text{intercept} \times 252$). Funds with Alpha > 0 generate excess returns above what market exposure predicts.

- **SBI Small Cap Fund - Regular Plan - Growth:** Alpha=30.34%, Beta=-0.023
- **DSP Small Cap Fund - Regular - Growth:** Alpha=30.06%, Beta=0.011
- **ICICI Pru Midcap Fund - Regular - Growth:** Alpha=29.26%, Beta=0.001
- **Mirae Asset Tax Saver Fund - Regular - Growth:** Alpha=28.27%, Beta=0.018
- **Kotak Flexicap Fund - Regular - Growth:** Alpha=27.33%, Beta=-0.023

5.3 VaR & CVaR (95% Historical)

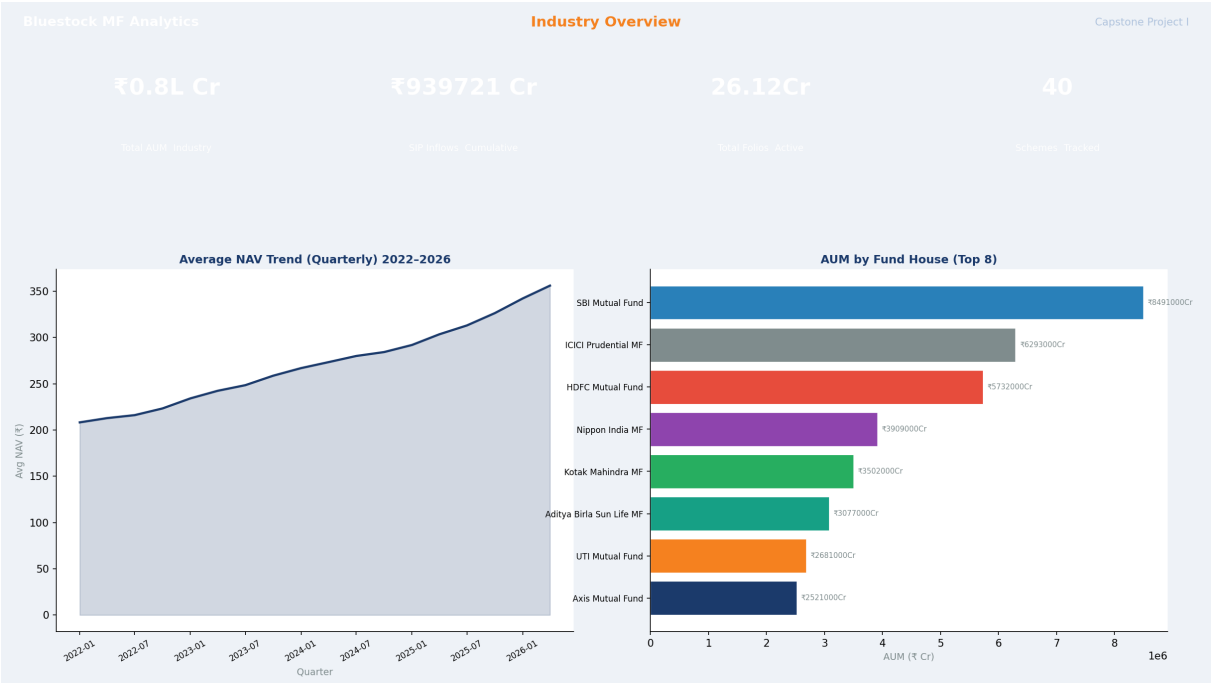
Historical VaR (95%) represents the 5th percentile of daily return distribution — the worst expected daily loss 95% of the time. CVaR (Conditional VaR) is the mean loss on days worse than VaR.

- Nippon India Gilt Securities Fund - Regular - Growth: VaR=-0.33%, CVaR=-0.45%
- HDFC Short Term Debt Fund - Regular - Growth: VaR=-0.33%, CVaR=-0.46%
- ABSL Liquid Fund - Regular - Growth: VaR=-0.02%, CVaR=-0.04%
- Kotak Liquid Fund - Regular - Growth: VaR=-0.02%, CVaR=-0.04%
- ICICI Pru Liquid Fund - Regular - Growth: VaR=-0.02%, CVaR=-0.03%

6. Dashboard Screenshots

Four dashboard pages were generated providing a visual analytical summary of the mutual fund universe.

Page 1: Industry Overview



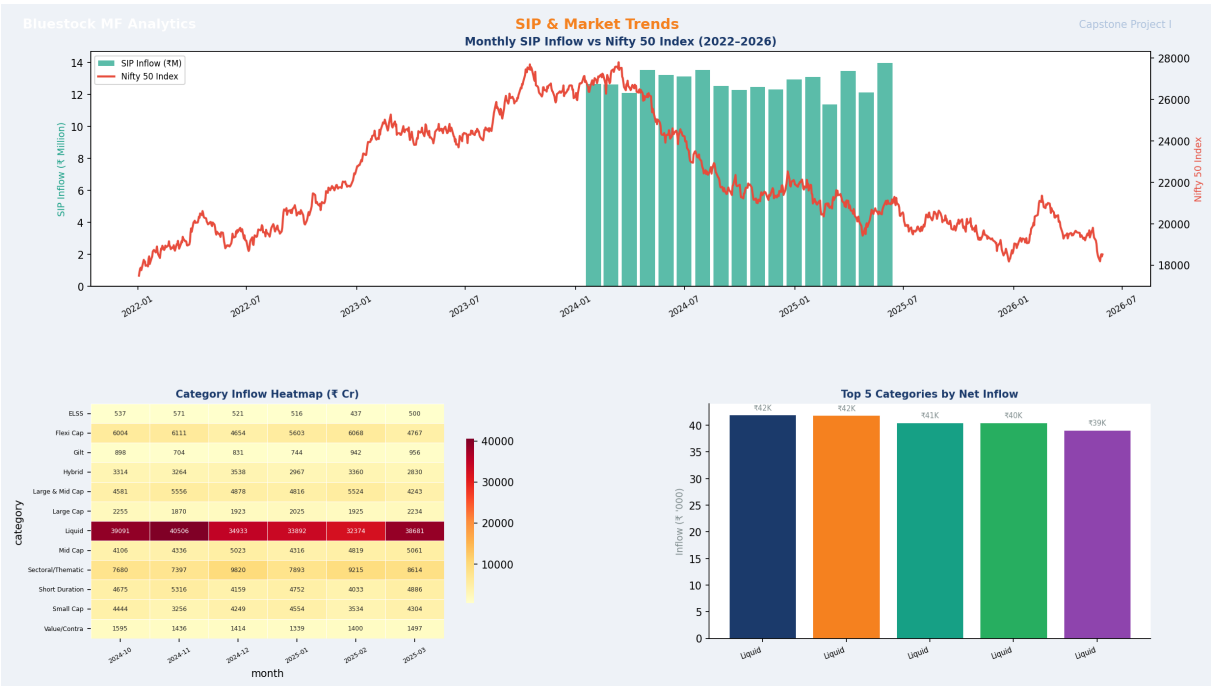
Page 2: Fund Performance



Page 3: Investor Analytics



Page 4: SIP & Market Trends



7. Limitations

- Dataset coverage is limited to Jan 2022 – May 2026 (~4.4 years), so 5-year CAGR uses maximum available history instead of a full 5-year window.
- Investor transaction data is synthetic (generated for demonstration), not real investor records.
- Portfolio holdings data covers only a subset of equity funds and represents a single snapshot date.
- Benchmark data (NIFTY 50, NIFTY 100) is used as proxy; sector-specific benchmarks for debt/hybrid funds are not included.
- The fund recommender uses Sharpe ratio as the sole ranking criterion within each risk bucket; a production system would incorporate multi-factor scoring.
- Power BI dashboard is represented by Python-generated matplotlib exports due to environment constraints; a real .pbix file would be required for full interactivity.

8. Recommendations

- Expand NAV history to 10+ years for statistically robust 5-year and 10-year performance comparisons.
- Integrate real-time AMFI data feed to keep NAVs current and enable live dashboard updates.
- Develop a multi-factor fund recommender incorporating Alpha, Max Drawdown, and Tracking Error in addition to Sharpe ratio.
- Implement a Streamlit or Dash web dashboard for interactive exploration by non-technical stakeholders.
- Apply machine learning (Random Forest or XGBoost) to predict future fund performance based on historical patterns and macroeconomic indicators.
- Conduct deeper SIP continuity analysis with investor segmentation to target at-risk investors with personalised nudges.
- Include international fund and Index ETF comparison to benchmark active vs passive strategies.

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